

A Minimality-Preserving Statman Translation

Yoshiki Nakamura

Naosuke Matsuda

Abstract

A formula is minimal in a logic L if it belongs to L and no strictly more general formula, in the sense of the substitution preorder, belongs to L . We show that the polarized extension-variable translation S of Tseitin and Statman preserves minimality: for every logic L that contains the axiom $I = p \rightarrow p$ and $S(\alpha)$, if α is minimal in L then so is $S(\alpha)$. This applies to all implicational logics containing **BCI**, to extensions of **FL_e** in the multiplicative language, and to extensions of **FL_{ew}** (in particular of intuitionistic logic) in the full language, and in these cases $L + \alpha = L + S(\alpha)$. For implicative α , $S(\alpha)$ has order at most four. Consequently, every implicative logic over **BCI** axiomatizable by formulas minimal in classical logic is axiomatizable by such formulas of order at most four, and four is optimal. Applied to the counter-example of Nakamura and Matsuda to the Komori–Kashima problem, this yields an explicit order-four axiom of Gödel–Dummett logic minimal in classical logic, answering their Open Problem 2. We also show that the base logic **BCI** cannot be weakened to **CK** or **BK**. The implicative results are formalized in Lean 4.

Keywords: Intermediate logic, Substructural logic, Minimal formulas, Substitution preorder, Axiomatization, Extension variables, Statman translation.

1 Introduction

1.1 The main theorem

Komori [13] studied logics through the *substitution preorder*: $\beta \leq_{\text{sub}} \alpha$ iff α is a substitution instance of β . A formula α is *minimal* in a logic L if $\alpha \in L$ and no strictly more general formula belongs to L . For example, Peirce’s law $((a \rightarrow b) \rightarrow a) \rightarrow a$ is minimal in classical logic **CL**.

The *polarized Statman translation* S (Definition 3.1) replaces every compound subformula occurrence u of α by a fresh variable x_u . It records the meaning of x_u in a premise E_u , a one-way definition whose direction depends on the polarity of u :

$$S(\alpha) = E_{u_1} \rightarrow \cdots \rightarrow E_{u_m} \rightarrow x_{\text{root}}.$$

For Peirce’s law, with x, y, w standing for the whole formula, $(a \rightarrow b) \rightarrow a$ and $a \rightarrow b$,

$$S(((a \rightarrow b) \rightarrow a) \rightarrow a) = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (w \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Each premise E_u contains exactly two connectives. So for implicative α , $S(\alpha)$ has order at most 4 and is computed in linear time.

Theorem 1.1 (Main Theorem). *Let L be a logic containing the identity axiom $I = p \rightarrow p$. If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

The hypothesis $S(\alpha) \in L$ holds for every $\alpha \in L$ as soon as L contains linear implicational logic **BCI** (Corollary 1.2 below). The axiom I cannot be dropped: there is a logic containing **BC**, but not I , in which the conclusion fails (Proposition 6.1).

1.2 Why the theorem holds

A generalization of $S(\alpha)$ can only “forget” some identifications made in $S(\alpha)$. The point is that each such forgetting can be pushed back to α . The proof has two steps.

1. *It suffices to consider one-step generalizations* (Corollary 5.5). A formula in L is minimal iff no formula obtained from it by replacing some occurrences of *one* subformula by a fresh variable z is in L .
2. *Every one-step generalization of $S(\alpha)$ decodes to one of α* (Lemma 5.8). Let α' be such a generalization. We give a substitution θ that maps each (possibly damaged) definition in α' to an instance $\gamma \rightarrow \gamma$ of I , and maps the conclusion to a one-step generalization β of α :

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \cdots \rightarrow (\gamma_k \rightarrow \gamma_k) \rightarrow \beta.$$

If $\alpha' \in L$, then $\beta \in L$ by modus ponens. This is impossible, because α is minimal.

Only closure under substitution and modus ponens and the axiom I are used. So the argument works for any logic. Polarity is needed only to guarantee $S(\alpha) \in L$. For Peirce’s law, a generalization of S that separates the two occurrences of w is decoded in Example 5.9.

1.3 What follows

Corollary 1.2. *Suppose that one of the following holds:*

- (i) *L is an implicational logic containing **BCI** (e.g. **BCI**, **BCK**, **IL**, **LC**, **CL**);*
- (ii) *L is a logic in the language $\{\rightarrow, \cdot, 0, 1\}$ containing the full Lambek calculus with exchange **FL_e**;*
- (iii) *L contains **FL_{ew}** (**FL_e** with weakening), in a language with $\rightarrow$ and some of $\cdot, \wedge, \vee, 0, 1, \perp, \top$ (e.g. **IL** in the full language).*

Then $L + \alpha = L + S(\alpha)$ for every α . If moreover α is minimal in L , then $S(\alpha)$ is minimal in L .

Taking $L = \mathbf{CL}$ for minimality and a base logic $L_0 \supseteq \mathbf{BCI}$ for the axiomatization, we obtain a normal form:

Corollary 1.3 (Order-four normal form). *Let L_0 be an implicational logic containing **BCI** (e.g. **BCI**, **BCK**, **IL**). If $L = L_0 + \Gamma$ with every $\alpha \in \Gamma$ minimal in **CL**, then $L = L_0 + \{S(\alpha) \mid \alpha \in \Gamma\}$, where every $S(\alpha)$ is minimal in **CL** and of order at most 4.*

For the Komori–Kashima problem [13, 12], this has the following consequences.

- Nakamura and Matsuda [14] gave a **CL**-minimal axiom G' of Gödel–Dummett logic **LC** of order 5, and asked whether order 4 suffices [14, Open Problem 2]. $S(G')$ is a **CL**-minimal axiom of **LC** of order 4, so the answer is yes. The formula is displayed in Corollary 4.2.
- The bound 4 is optimal: classical tautologies of order at most 3 are intuitionistic theorems (Proposition 4.1).
- Corollary 1.3 holds over **BCK**, so it also applies to Komori’s original super-**BCK** formulation [14, footnote 1]. For the question which logics are axiomatizable by **CL**-minimal formulas [14, Open Problem 1], one may restrict attention to order 4.

The base logic **BCI** cannot be weakened by dropping B , C or I (Propositions 6.1 and 6.2). The implicational case of the main theorem and Corollary 1.2(i) are formalized in Lean 4 (Section 7).

1.4 Related work and outline

The translation is not new. Extension variables go back to Tseitin [9]. Statman [15] used them to reduce intuitionistic propositional logic to its implicational fragment (see also Švejdar [8]). One-way definitions chosen by polarity appear in the Plaisted–Greenbaum translation [6] and in Mints’ normal form [5]. Rybakov [7] uses extension variables for rules. Closest to us is Jeřábek [4]. His φ^+ is our $S(\varphi)$ with an extra “ $\wedge 1$ ” in each definition; with it he shows that the substructural hierarchy of Ciabattoni, Galatos and Terui [2] collapses to $\mathcal{N}_3$ over **FL_e**. To our knowledge, the preservation of minimality has not been observed before. The extra “ $\wedge 1$ ” destroys it (Section 6.2).

[\[Plot: Check the exact forms used in \[15\], \[8\], \[5\].\]](#)

The rest of the paper is organized as follows.

- Section 2: preliminaries.
- Section 3: the translation and its basic properties.
- Section 4: the consequences above, with $S(G')$ displayed.
- Section 5: the proof of Theorem 1.1.
- Section 6: supplementary remarks (weaker base logics, additive connectives, premises of order 3).
- Section 7: the formalization.
- Section 8: open problems.

2 Preliminaries

Formulas and logics. We work in a propositional language $\mathcal{L}$ that contains $\rightarrow$ and possibly some of the binary connectives $\cdot$ (fusion), $\wedge$, $\vee$ and some constants (among $\perp$, $\top$, 0 , 1). An *atom* is a variable or a constant. $\alpha \rightarrow \beta \rightarrow \gamma$ stands for $\alpha \rightarrow (\beta \rightarrow \gamma)$. As in [14, Sect. 2],

a *logic* is a set L of $\mathcal{L}$ -formulas closed under modus ponens and substitution; $L + F$ is the smallest logic containing $L \cup F$. L is *consistent* if it is not the set of all formulas. Since L is closed under substitution, L is consistent iff no variable belongs to L .

We use the following logics (as sets of theorems of their usual Hilbert calculi, see [3]):

- **BCI**, **BCK**, **IL** and **CL** in the implicative language. $\mathbf{IL} = \{K, S\}^*$ and $\mathbf{CL} = \mathbf{IL} + P$ as in [14].
- **FL_e** (full Lambek calculus with exchange) and **FL_{ew}** (with exchange and weakening) in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$, possibly with $\perp, \top$. Intuitionistic logic in the full language contains **FL_{ew}**.

For implicative formulas, $\alpha \in \mathbf{CL}$ iff α is true under every Boolean valuation.

A *substitution* σ maps variables to formulas; $\sigma\alpha$ is the result of applying it to α . We write $\beta \leq_{\text{sub}} \alpha$ if $\alpha = \sigma\beta$ for some σ , and $\beta <_{\text{sub}} \alpha$ if $\beta \leq_{\text{sub}} \alpha$ but not $\alpha \leq_{\text{sub}} \beta$. A formula α is *minimal in L* if $\alpha \in L$ and there is no $\beta \in L$ with $\beta <_{\text{sub}} \alpha$. The *order* of an implicative formula is given by $\text{Ord}(p) = 1$ and $\text{Ord}(\alpha \rightarrow \beta) = \max\{\text{Ord}(\alpha) + 1, \text{Ord}(\beta)\}$.

Every logic L satisfies (C1) and (C2), and (C3) holds whenever L contains I :

- (C1) if $\gamma \in L$, then $\sigma\gamma \in L$ for every substitution σ ;
- (C2) if $\gamma_1 \rightarrow \dots \rightarrow \gamma_k \rightarrow \beta \in L$ and $\gamma_1, \dots, \gamma_k \in L$, then $\beta \in L$.
- (C3) if L contains the identity axiom $I = p \rightarrow p$, then $\gamma \rightarrow \gamma \in L$ for every formula γ , by (C1).

Occurrences. We identify an occurrence in a formula α with its position in the syntax tree: a finite word over $\{0, 1\}$. The empty word ε is the root. If the subformula at u is $\gamma \circ \delta$ ($\circ$ a binary connective), then γ is at $u0$ and δ is at $u1$. We write α_u for the subformula at u . We say that u is *strictly below* v if v is a proper prefix of u . An occurrence u is a *leaf* if α_u is an atom, and a *compound occurrence* otherwise. Two occurrences of the same formula are never strictly below one another.

For a set Q of occurrences in α , no one of which is strictly below another, and a variable z , let $\alpha[Q := z]$ be the formula obtained from α by replacing the subformula at each $u \in Q$ by z .

3 The Polarized Statman Translation

Fix a formula α . Assign a *polarity* to each occurrence of α : ε is positive. If u is a compound occurrence with $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$, then $u1$ has the polarity of u and $u0$ the opposite one. If $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ with $\circ \in \{\cdot, \wedge, \vee\}$, both $u0$ and $u1$ have the polarity of u . For each compound occurrence u choose a new variable x_u (an *extension variable*), distinct from the variables of α and from each other. For every occurrence u put

$$r_u = \begin{cases} \alpha_u, & u \text{ a leaf,} \\ x_u, & u \text{ compound,} \end{cases}$$

and for every compound occurrence u with $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ put $b_u = r_{u0} \circ r_{u1}$ (the *body*) and

$$E_u = \begin{cases} b_u \rightarrow x_u, & u \text{ positive,} \\ x_u \rightarrow b_u, & u \text{ negative} \end{cases} \quad (\text{the link of } u).$$

Definition 3.1. Let $u_1, \dots, u_m$ be the compound occurrences of α in post-order. Then $S(\alpha) = E_{u_1} \rightarrow \dots \rightarrow E_{u_m} \rightarrow r_\varepsilon$.

For implicative α , $S(\alpha)$ is implicative. Jeřábek's φ^+ [4, Def. 3.6] uses $(b_u \rightarrow x_u) \wedge 1$ in place of E_u and a product in place of the iterated implication. We explain in Remark 6.3 why the “ $\wedge 1$ ” must be dropped here.

Example 3.2. Let $\alpha = ((a \rightarrow b) \rightarrow a) \rightarrow a$ (Peirce's law). Its compound occurrences are ε (positive), 0 (negative) and 00 (positive), with $\alpha_{00} = a \rightarrow b$ and $\alpha_0 = (a \rightarrow b) \rightarrow a$. Writing x, y, w for $x_\varepsilon, x_0, x_{00}$:

$$S(\alpha) = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (w \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Lemma 3.3. If α is implicative, $S(\alpha)$ has $3m + 1$ variable occurrences and $\text{Ord}(S(\alpha)) \leq 4$.

Proof. Each E_u has three variable occurrences. $\text{Ord}(b_u) = 2$, so $\text{Ord}(b_u \rightarrow x_u) = 3$ and $\text{Ord}(x_u \rightarrow b_u) = 2$. Hence $\text{Ord}(S(\alpha)) \leq \max\{3 + 1, 1\} = 4$. $\square$

The next lemma lists the facts about the shape of $S(\alpha)$ used below. All are immediate from the definition.

Lemma 3.4 (Soundness). *In each of the following cases, $L \vdash \alpha \rightarrow S(\alpha)$ for every formula α of the given language:*

- (i) $\mathcal{L} = \{\rightarrow\}$ and $L \supseteq \mathbf{BCI}$;
- (ii) $\mathcal{L} \subseteq \{\rightarrow, \cdot, 0, 1\}$ and $L \supseteq \mathbf{FL}_e$;
- (iii) $L \supseteq \mathbf{FL}_{ew}$, in any sublanguage of $\{\rightarrow, \cdot, \wedge, \vee, 0, 1, \perp, \top\}$ containing $\rightarrow$; in particular $L \supseteq \mathbf{IL}$.

Proof. We argue in the sequent calculus for $\mathbf{FL}_e$ (with weakening in case (iii)). For a compound occurrence u , let Γ_u be the multiset of the links E_v with $v = u$ or v strictly below u ; for a leaf, Γ_u is empty. We show by induction from the leaves upward:

$$\Gamma_u, \alpha_u \vdash r_u \text{ (} u \text{ positive),} \quad \Gamma_u, r_u \vdash \alpha_u \text{ (} u \text{ negative).}$$

For leaves this is an axiom. Let u be positive with $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$. The occurrence $u0$ is negative and $u1$ is positive. From the induction hypotheses and $\rightarrow$ -left, $\Gamma_{u0}, \Gamma_{u1}, \alpha_u, r_{u0} \vdash r_{u1}$. By $\rightarrow$ -right and $\rightarrow$ -left with $E_u = (r_{u0} \rightarrow r_{u1}) \rightarrow x_u$, we get $\Gamma_u, \alpha_u \vdash x_u$. The case of fusion is similar. Every formula of Γ_u is used exactly once, so no structural rule other than exchange is needed; this proves (i) and (ii). For $\wedge$ and $\vee$, the right- $\wedge$ and left- $\vee$ rules share the context. In each branch we remove the links of the other subtree by weakening; this proves (iii). Negative occurrences are symmetric. Taking $u = \varepsilon$ gives $E_{u_1}, \dots, E_{u_m}, \alpha \vdash r_\varepsilon$, i.e. $\vdash \alpha \rightarrow S(\alpha)$ after exchange. $\square$

Lemma 3.5 (Decoding). *Let θ_α be the substitution with $\theta_\alpha(x_u) = \alpha_u$ for each compound occurrence u , fixing all other variables. Then $\theta_\alpha(r_u) = \alpha_u$ for all u , and*

$$\theta_\alpha S(\alpha) = (\alpha_{u_1} \rightarrow \alpha_{u_1}) \rightarrow \cdots \rightarrow (\alpha_{u_m} \rightarrow \alpha_{u_m}) \rightarrow \alpha.$$

Consequently, if a logic L contains I , then $\alpha \in L + S(\alpha)$, and $S(\alpha) \in L$ implies $\alpha \in L$.

Proof. $\theta_\alpha(r_u) = \alpha_u$ holds by definition. Hence $\theta_\alpha(b_u) = \alpha_{u_0} \circ \alpha_{u_1} = \alpha_u = \theta_\alpha(x_u)$, so $\theta_\alpha(E_u) = \alpha_u \rightarrow \alpha_u$ in both polarities. The rest follows from (C1)–(C3). $\square$

Corollary 3.6. *In each case of Lemma 3.4, $L + \alpha = L + S(\alpha)$.*

4 Consequences for the Komori–Kashima Problem

In this section we derive the consequences stated in the introduction, assuming Theorem 1.1, which is proved in Section 5. We work in the implicative language.

Proof of Corollary 1.2. By Lemma 3.4, $L \vdash \alpha \rightarrow S(\alpha)$, so $S(\alpha) \in L$ whenever $\alpha \in L$. Apply Theorem 1.1. The equation $L + \alpha = L + S(\alpha)$ is Corollary 3.6. $\square$

Proof of Corollary 1.3. By Corollary 1.2(i) for L_0 , $L_0 + \alpha = L_0 + S(\alpha)$ for each $\alpha \in \Gamma$. By Theorem 1.1 with $L = \mathbf{CL}$ (where $S(\alpha) \in \mathbf{CL}$ by Lemma 3.4), each $S(\alpha)$ is minimal in $\mathbf{CL}$. It has order at most 4 by Lemma 3.3. $\square$

Proposition 4.1 (Four is optimal). *Every implicative formula in $\mathbf{CL}$ of order at most 3 is in $\mathbf{IL}$.*

Proof. A formula of order ≤ 2 has the form $p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$ with variables p_i, q ; read it as the rule “from $p_1, \dots, p_k$ infer q ”. A formula of order ≤ 3 is $F = H_1 \rightarrow \cdots \rightarrow H_n \rightarrow r$ with such H_i and a variable r . Let C be the closure of $\emptyset$ under the rules $H_1, \dots, H_n$. The valuation making exactly C true satisfies every H_i , so $r \in C$ because $F \in \mathbf{CL}$. Following the closure step by step, modus ponens derives r from $H_1, \dots, H_n$ in $\mathbf{IL}$. Hence $F \in \mathbf{IL}$. $\square$

Corollary 4.2 (Answer to [14, Open Problem 2]). *Let G' be the formula of [14, Sect. 4]:*

$$\begin{aligned} G' = & (a \rightarrow a') \rightarrow (b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \\ & \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c. \end{aligned}$$

Then $S(G')$ is the formula

$$\begin{aligned} S(G') = & (x_2 \rightarrow a \rightarrow a') \rightarrow (x_4 \rightarrow b' \rightarrow b) \rightarrow (x_6 \rightarrow c'' \rightarrow c') \rightarrow (x_8 \rightarrow c''' \rightarrow c') \\ & \rightarrow ((a \rightarrow b) \rightarrow x_{11}) \rightarrow (x_{10} \rightarrow x_{11} \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow x_{14}) \rightarrow (x_{13} \rightarrow x_{14} \rightarrow c''') \\ & \rightarrow (x_{18} \rightarrow a' \rightarrow b') \rightarrow ((x_{18} \rightarrow c'') \rightarrow x_{17}) \rightarrow (x_{21} \rightarrow b \rightarrow a) \rightarrow ((x_{21} \rightarrow c''') \rightarrow x_{20}) \\ & \rightarrow (x_{22} \rightarrow c' \rightarrow c) \rightarrow (x_{19} \rightarrow x_{20} \rightarrow x_{22}) \rightarrow (x_{16} \rightarrow x_{17} \rightarrow x_{19}) \rightarrow ((x_{16} \rightarrow c) \rightarrow x_{15}) \\ & \rightarrow ((x_{13} \rightarrow x_{15}) \rightarrow x_{12}) \rightarrow ((x_{10} \rightarrow x_{12}) \rightarrow x_9) \rightarrow ((x_8 \rightarrow x_9) \rightarrow x_7) \\ & \rightarrow ((x_6 \rightarrow x_7) \rightarrow x_5) \rightarrow ((x_4 \rightarrow x_5) \rightarrow x_3) \rightarrow ((x_2 \rightarrow x_3) \rightarrow x_1) \rightarrow x_1. \end{aligned}$$

It has 22 premises, 30 variables and 67 variable occurrences. It satisfies $\mathbf{IL} + S(G') = \mathbf{LC}$, it is minimal in $\mathbf{CL}$, and $\text{Ord}(S(G')) = 4$. Hence there exists a proper intermediate logic

axiomatizable by a formula minimal in **CL** of order at most 4. That is, [14, Open Problem 2] has an affirmative answer, and the Komori–Kashima conjecture already fails for formulas of order at most 4.

Proof. By [14, Lemma 4.2], $\mathbf{IL} + G' = \mathbf{LC}$ and G' is minimal in **CL**. Now apply Theorem 1.1, Corollary 3.6 and Lemma 3.3. The displayed formula is Definition 3.1, with the implication occurrences numbered as in Appendix A. $\square$

[Plot: Table 1 lists the subformula of G' that each x_i stands for. The Lean file checks that the displayed formula equals `statman Gprime 8` and has order 4.]

5 Proof of the Main Theorem

We follow the two steps described in the introduction. Section 5.1 proves the criterion (step 1). Section 5.2 proves the decoding (step 2) and Theorem 1.1.

5.1 A criterion for minimality

Definition 5.1. Let δ be a formula, let Q be a nonempty set of occurrences of δ in α (that is, $\alpha_u = \delta$ for $u \in Q$), and let z be a variable not occurring in α . If δ is a variable, assume moreover that some occurrence of δ in α is not in Q . Then $\alpha[Q := z]$ is called a *one-step generalization* of α .

Example 5.2. For $\alpha = (a \rightarrow b) \rightarrow (a \rightarrow b)$, both $z \rightarrow (a \rightarrow b)$ (with $\delta = a \rightarrow b$) and $(z \rightarrow b) \rightarrow (a \rightarrow b)$ (with $\delta = a$) are one-step generalizations of α . $(z \rightarrow b) \rightarrow (z \rightarrow b)$ is not one, because all occurrences of a are replaced.

Lemma 5.3. *Every one-step generalization $\beta = \alpha[Q := z]$ of α satisfies $\beta <_{\text{sub}} \alpha$.*

Proof. $\beta \leq_{\text{sub}} \alpha$, because substituting δ for z in β gives α (z does not occur in α). Suppose, for a contradiction, that $\tau\alpha = \beta$ for some τ . Call connectives and constants *symbols*. A substitution never decreases the number of occurrences of symbols.

If δ is not a variable, it contains a symbol, so β has fewer symbols than α . This is a contradiction.

If δ is a variable p , then α and β have the same number of symbols, so τ maps each variable of α to a variable. Then $\tau\alpha$ has the same syntax tree as α , with the variable $\tau(p)$ at every leaf where α has p . But at these leaves β has z (at $u \in Q$) and p (at the occurrence outside Q). Hence $\tau(p) = z$ and $\tau(p) = p$, a contradiction. $\square$

Lemma 5.4. *Let L be a logic and $\alpha \in L$. If no one-step generalization of α is in L , then α is minimal in L .*

Proof. Let $\gamma \in L$ and $\sigma\gamma = \alpha$. We show $\alpha \leq_{\text{sub}} \gamma$.

Suppose first that σ maps the variables of γ injectively to variables. Then there is a substitution τ with $\tau(\sigma(q)) = q$ for every variable q of γ , and $\tau\alpha = \tau\sigma\gamma = \gamma$.

Otherwise, there is a variable q of γ such that either (i) $\sigma(q)$ is not a variable, or (ii) $\sigma(q) = \sigma(q')$ is a variable for some variable $q' \neq q$ of γ . Let z be a variable not occurring in α , and let σ' agree with σ except that $\sigma'(q) = z$. Every occurrence of q in γ yields an

occurrence of $\delta := \sigma(q)$ in $\alpha = \sigma\gamma$. Let Q be the set of these occurrences; $Q \neq \emptyset$. Then $\sigma'\gamma = \alpha[Q := z]$. In case (ii), the occurrences of q' in γ yield occurrences of the variable δ that are not in Q . So $\sigma'\gamma$ is a one-step generalization of α , and $\sigma'\gamma \in L$ by (C1). This contradicts the hypothesis. $\square$

Corollary 5.5. *Let L be a logic. A formula $\alpha \in L$ is minimal in L if and only if no one-step generalization of α is in L .*

Proof. “If” is Lemma 5.4. “Only if” follows from Lemma 5.3. $\square$

5.2 Decoding one-step generalizations

For an occurrence u of α , the *reading* of u is the occurrence of r_u inside the body b_v of its parent v , or the last formula r_ε of $S(\alpha)$ when $u = \varepsilon$. The next lemma lists the facts about the shape of $S(\alpha)$ that we use. All of them are immediate from the definition.

Lemma 5.6. (a) *For each compound occurrence u , x_u occurs in $S(\alpha)$ exactly twice: at the reading of u , and as the head of E_u (the variable side of the link).*

(b) *The occurrences of atoms of α in $S(\alpha)$ are exactly the readings of the leaves of α . So for each atom c they correspond bijectively to the leaves u with $\alpha_u = c$.*

(c) *Every compound subformula occurrence in $S(\alpha)$ is of exactly one of the following kinds:*

- *a body b_u (both immediate subformulas are atoms);*
- *a link E_u (one immediate subformula is an atom, the other a body);*
- *a tail $T_i = E_{u_i} \rightarrow \cdots \rightarrow E_{u_m} \rightarrow r_\varepsilon$ with $1 \leq i \leq m$ (its left immediate subformula is a link).*

In particular, no tail is a subformula of a link, and distinct tails are distinct formulas.

(d) *If $b_u = b_v$ as formulas, then $\alpha_u = \alpha_v$. If $E_u = E_v$ as formulas, then $u = v$.*

Proof. (a)–(c) follow by inspecting Definition 3.1. For the last claims of (c): the proper subformulas of a link are its body and atoms, and T_i has $m - i + 1$ premises. For (d), note that r_w determines α_w : if r_w is an atom of α , then $\alpha_w = r_w$; if $r_w = x_{w'}$, then $w = w'$. So $b_u = b_v$ gives the same main connective, $\alpha_{u0} = \alpha_{v0}$ and $\alpha_{u1} = \alpha_{v1}$, hence $\alpha_u = \alpha_v$. A positive link has the form (body) $\rightarrow$ (variable), and a negative one (variable) $\rightarrow$ (body). So $E_u = E_v$ forces equal polarity and $x_u = x_v$, hence $u = v$. $\square$

We call a one-step generalization $S(\alpha)[P := z]$ *normal* if every position in P is a reading or a body. We first reduce to normal ones. Throughout, z is a variable not occurring in $S(\alpha)$.

Lemma 5.7 (Normalization). *If some one-step generalization of $S(\alpha)$ belongs to a logic L , then some normal one-step generalization of $S(\alpha)$ belongs to L .*

Proof. Let $\alpha' = S(\alpha)[P := z] \in L$, where P is a set of occurrences of δ . We go through the kinds of δ given by Lemma 5.6. Each step replaces α' by a substitution instance, which stays in L by (C1).

- δ is an atom of α or a body. Then every occurrence of δ is a reading, resp. a body, by Lemma 5.6(b),(c). So α' is already normal.
- $\delta = x_u$. By Lemma 5.6(a) and the definition of a one-step generalization, P is one of the two occurrences of x_u . If it is the reading, α' is normal. If it is the head, exchange the variables x_u and z . The result is $S(\alpha)$ with the reading of u replaced by z , which is normal.
- $\delta = E_u$. By Lemma 5.6(c),(d), E_u occurs only once, so α' is $S(\alpha)$ with the premise E_u replaced by z . Substitute for z the link E_u with its head replaced by z . This gives $S(\alpha)$ with the head of E_u replaced by z , which is the previous case.
- $\delta = T_i$. By Lemma 5.6(c), T_i occurs only once, so $\alpha' = E_{u_1} \rightarrow \dots \rightarrow E_{u_{i-1}} \rightarrow z$. Substitute for z the tail T_i with its last formula r_ε replaced by z . This gives $S(\alpha)$ with the reading of ε replaced by z , which is normal. $\square$

Lemma 5.8 (Decoding). *Let $\alpha' = S(\alpha)[P := z]$ be a normal one-step generalization. Let Q be the set of occurrences u of α such that the reading or the body of u lies in P . Then $\beta = \alpha[Q := z]$ is a one-step generalization of α , and there is a substitution θ with*

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \dots \rightarrow (\gamma_k \rightarrow \gamma_k) \rightarrow \beta.$$

Proof. β is a one-step generalization. Q is nonempty. All α_u ($u \in Q$) are the same formula δ_0 :

- if δ is an atom, then $\delta_0 = \delta$;
- if $\delta = x_w$, then $Q = \{w\}$;
- if δ is a body, use Lemma 5.6(d).

In particular, no element of Q is strictly below another. If δ_0 is a variable, then $\delta = \delta_0$ and Q misses some leaf labelled δ_0 by Lemma 5.6(b).

The substitution. Call v *hidden* if it is strictly below some element of Q . For v not hidden, write β_v for the subformula of β at v ($\beta_v = z$ for $v \in Q$). Let θ fix z and the variables of α , and put

$$\theta(x_v) = \begin{cases} \alpha_v, & \text{if } v \text{ is hidden, or the reading of } v \text{ is in } P, \\ \beta_v, & \text{otherwise.} \end{cases}$$

Here is the rule behind this choice: x_v is sent to α_v when only its head survives, and to β_v when its reading survives.

Readings. If v is hidden, θ maps the reading of v to α_v . If v is not hidden, it maps it to β_v :

- if the reading is in P , it is $z = \beta_v$;
- otherwise it is r_v , and $\theta(r_v) = \beta_v$ by definition (for a leaf $v \notin Q$, $\beta_v = \alpha_v = r_v$).

(Positions below an element of Q are not in P , because Q has no nested elements.)

Premises. Let v be compound, with main connective $\circ$.

- If v is hidden, E_v is unchanged, and θ maps its body to $\alpha_{v0} \circ \alpha_{v1} = \alpha_v = \theta(x_v)$.
- If v is not hidden and $v \notin Q$, E_v is unchanged, and θ maps its body to $\beta_{v0} \circ \beta_{v1} = \beta_v = \theta(x_v)$.
- If $v \in Q$ and its body is in P , then E_v has become $z \rightarrow x_v$ or $x_v \rightarrow z$, and $\theta(x_v) = \beta_v = z$.
- If $v \in Q$ and its reading is in P , then E_v is unchanged. Its children are hidden, so θ maps its body to $\alpha_v = \theta(x_v)$.

In every case the premise is mapped to $\gamma \rightarrow \gamma$ (in either polarity).

Conclusion. The last formula is the reading of ε , which is not hidden. So it is mapped to $\beta_\varepsilon = \beta$. $\square$

Proof of Theorem 1.1. If α is an atom, then $S(\alpha) = \alpha$ and there is nothing to prove. Otherwise, suppose some one-step generalization of $S(\alpha)$ is in L . By Lemma 5.7, some normal one-step generalization α' is in L . By Lemma 5.8 and (C1)–(C3), the one-step generalization β of α is in L . This contradicts the minimality of α (Corollary 5.5). Hence no one-step generalization of $S(\alpha)$ is in L . Since $S(\alpha) \in L$, $S(\alpha)$ is minimal by Corollary 5.5. $\square$

Consistency of L is not needed. If α is compound and minimal in L , then the variable z , a one-step generalization of α , is not in L .

Example 5.9. We illustrate Lemma 5.8 with Example 3.2. Replace the reading of 00, i.e. the second occurrence of $w = x_{00}$, by z :

$$\alpha' = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (z \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Here $Q = \{00\}$ and $\beta = (z \rightarrow a) \rightarrow a$. The substitution is $w \mapsto a \rightarrow b$ (only the head of w survives), $y \mapsto z \rightarrow a$ and $x \mapsto \beta$:

$$\theta\alpha' = ((a \rightarrow b) \rightarrow (a \rightarrow b)) \rightarrow ((z \rightarrow a) \rightarrow (z \rightarrow a)) \rightarrow (\beta \rightarrow \beta) \rightarrow \beta.$$

Since $\beta \notin \mathbf{CL}$ ($z = 0$, $a = 0$), neither is α' .

6 Further Remarks

This section collects supplementary results. They are not needed for the main line of the paper.

6.1 How far can the base logic be weakened?

Theorem 1.1 needs only $I \in L$ and $S(\alpha) \in L$. In the implicational case, the last condition is guaranteed by **BCI** (Lemma 3.4). We show that none of I , B , C can be dropped. We write **XY**... for the implicational logic axiomatized by the axiom schemes $X, Y, \dots$ among

$$B = (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, \quad B' = (a \rightarrow b) \rightarrow (b \rightarrow c) \rightarrow a \rightarrow c, \quad C = (a \rightarrow b \rightarrow c) \rightarrow b \rightarrow a \rightarrow c,$$

$K = a \rightarrow b \rightarrow a$ and $I = a \rightarrow a$, with modus ponens and substitution.

The axiom I cannot be dropped. The proof uses I only to discharge the premises $\gamma \rightarrow \gamma$ produced by the decoding (Lemma 5.8). The next proposition shows that this use is essential, already over **BC**.

Proposition 6.1. *Let $M = (\{0, 1, 2\}, \rightarrow, \{0\})$ with*

$\rightarrow$	0	1	2
0	0	1	1
1	0	0	0
2	0	1	1

*and let L_M be the set of formulas that take the value 0 under every valuation in M . Then L_M is a logic containing **BC** but not I . Let $\alpha = (p \rightarrow q) \rightarrow (p \rightarrow q)$. Then:*

- (a) α is minimal in L_M ;
- (b) $S(\alpha) = (x_1 \rightarrow (p \rightarrow q)) \rightarrow ((p \rightarrow q) \rightarrow x_2) \rightarrow ((x_1 \rightarrow x_2) \rightarrow x_0) \rightarrow x_0 \in L_M$;
- (c) $S(\alpha)$ is not minimal in L_M : its one-step generalization

$$\alpha' = (x_1 \rightarrow z) \rightarrow (z \rightarrow x_2) \rightarrow ((x_1 \rightarrow x_2) \rightarrow x_0) \rightarrow x_0,$$

obtained by replacing both bodies $p \rightarrow q$ by z , belongs to L_M .

So Theorem 1.1 fails for logics not containing I , even when $S(\alpha) \in L$.

Proof. L_M is closed under substitution. It is closed under modus ponens because $0 \rightarrow y = 0$ forces $y = 0$. A finite check shows that B and C take only the value 0, while $2 \rightarrow 2 = 1$, so $I \notin L_M$.

(a) We have $\alpha \in L_M$: the values of implications lie in $\{0, 1\}$, and $0 \rightarrow 0 = 1 \rightarrow 1 = 0$. The one-step generalizations of α are

$$(z \rightarrow q) \rightarrow (p \rightarrow q), \quad (p \rightarrow z) \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow (z \rightarrow q), \quad (p \rightarrow q) \rightarrow (p \rightarrow z),$$

$$z \rightarrow (p \rightarrow q), \quad (p \rightarrow q) \rightarrow z, \quad z \rightarrow z, \quad z.$$

Each of them takes a value $\neq 0$ under some valuation in M . For $z \rightarrow z$ take $z = 2$, and for z take $z = 1$. For each of the remaining six, a refuting valuation is found by a finite check.

(b) and (c) are finite checks as well. In fact α' is the type of the linear term $\lambda fgh. h(\lambda y. g(fy))$, and $S(\alpha)$ is its instance under $z \mapsto p \rightarrow q$. Both are theorems of **BC**. $\square$

Decoding α' as in Lemma 5.8 gives $\beta = z \rightarrow z$. The premise that would have to be discharged is exactly $z \rightarrow z$, which is not available in L_M . Validity of α provides $X \rightarrow X$ only for formulas X whose value is that of an implication.

[Plot: Checks: [tools/bc_alpha_matrix.py](#) (matrix M , found with **z3** and verified by exhaustive evaluation) and [tools/bc_check.py](#) (condensed detachment: $S(\alpha), \alpha' \in \mathbf{BC}$).]

Proposition 6.2. (a) (Without B .) K is minimal in **CK**, but $S(K) = ((b \rightarrow a) \rightarrow x_1) \rightarrow ((a \rightarrow x_1) \rightarrow x_0) \rightarrow x_0 \notin \mathbf{CK}$.

(b) (Without C .) I is minimal in $\mathbf{BB'KI}$, but $S(I) = ((a \rightarrow a) \rightarrow x) \rightarrow x \notin \mathbf{BB'KI}$.
Likewise K is minimal in $\mathbf{BK}$ and in $\mathbf{BB'KI}$, but $S(K) \notin \mathbf{BB'KI} \supseteq \mathbf{BK}$.

Hence, over $\mathbf{CK}$, $\mathbf{BK}$ and $\mathbf{BB'KI}$ (and every logic between $\mathbf{BK}$ and $\mathbf{BB'KI}$), S does not preserve minimality, and $L + \alpha \neq L + S(\alpha)$ for $\alpha = K$ (resp. I).

Proof. Minimality. The one-step generalizations of K are $z \rightarrow b \rightarrow a$, $a \rightarrow b \rightarrow z$, $a \rightarrow z$ and z ; those of I are $z \rightarrow a$, $a \rightarrow z$ and z . None of them is classically valid, so none lies in any of the logics considered, all of which are contained in $\mathbf{CL}$.

(a) Let $M_4 = (\{0, 1, 2, 3\}, \rightarrow, \{0\})$ with the table

$\rightarrow$	0	1	2	3
0	0	1	2	3
1	0	0	1	0
2	0	0	0	3
3	0	1	2	0

A finite check shows that M_4 validates C and K (hence also I , which is derivable in $\mathbf{CK}$), but not B . The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ forces $y = 0$. So every $\mathbf{CK}$ -theorem takes the value 0. With $a = b = 1$, $x_1 = 2$ and $x_0 = 3$, however, $b \rightarrow a = 0$, $(b \rightarrow a) \rightarrow x_1 = 2$, $a \rightarrow x_1 = 1$ and $(a \rightarrow x_1) \rightarrow x_0 = 0$. Hence $S(K)$ takes the value $2 \rightarrow (0 \rightarrow 3) = 2 \rightarrow 3 = 3 \neq 0$. Intuitively, $S(K)$ needs the composition $\lambda y. e_1(\lambda z. y)$ inside an argument, which the combinators C, K cannot form. By contrast, $S(I) = ((a \rightarrow a) \rightarrow x) \rightarrow x$ is a $\mathbf{CK}$ -theorem (it is the type of CII).

(b) Let $M_3 = (\{0, 1, 2\}, \rightarrow, \{0\})$ with $x \rightarrow y = 0$ if $y \leq x$ and $x \rightarrow y = 1$ otherwise. M_3 validates B, B', K, I : this is a finite check. The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ forces $y = 0$. So every theorem of $\mathbf{BB'KI}$ takes the value 0 in M_3 . With $a = 0$ and $x = x_0 = 2$, however, $S(I)$ takes the value $(0 \rightarrow 2) \rightarrow 2 = 1 \rightarrow 2 = 1$. $S(K)$ is refuted similarly. (M_3 refutes C and W .) $\square$

[Plot: Checks: `tools/ck_z3.py` (matrix M_4 , found with `z3` and verified by exhaustive evaluation), `tools/bki_matrix.py` (matrix M_3), `tools/ck_cd.py` and `tools/bk_cd.py` (condensed-detachment search, consistent with (a) and (b)). Open: ticket entailment $\mathbf{T}_{\rightarrow} = \mathbf{BB'IW}$, where M_3 does not apply because it refutes W ; and whether reordering the links helps. For (b), every order of links fails already for $S(I)$, which has a single link.]

6.2 Additive connectives over $\mathbf{FL}_e$

Remark 6.3 (Additive connectives over $\mathbf{FL}_e$). The proof of Theorem 1.1 uses only (C1)–(C3), with $I \in L$. Polarity is used only in Lemma 3.4, which is where $\mathbf{FL}_e$ with $\wedge, \vee$ fails. Two facts, both checked with a cut-free prover for $\mathbf{FL}_e$ (`tools/fle_check.py`), show that the problem cannot be removed by simply adding Jeřábek’s “ $\wedge 1$ ”:

- (a) *Without wrappers, S need not be sound.* $\alpha = (q \rightarrow q) \wedge (r \rightarrow r)$ is minimal in $\mathbf{FL}_e$, but $S(\alpha) = ((q \rightarrow q) \rightarrow x_0) \rightarrow ((r \rightarrow r) \rightarrow x_1) \rightarrow ((x_0 \wedge x_1) \rightarrow x) \rightarrow x$ is not provable in $\mathbf{FL}_e$: each branch of the $\wedge$ -right rule would have to discard the other branch’s link.
- (b) *With wrappers, minimality is always lost.* Jeřábek’s φ^+ puts $(E_u \wedge 1)$ in place of E_u . The link of the root is never discarded, so replacing its 1 by z gives a one-step generalization

that is still provable (by $\wedge$ -left). Hence φ^+ is not minimal in any $L \supseteq \mathbf{FL}_e$ (including $\mathbf{FL}_{ew}$, $\mathbf{IL}$, $\mathbf{CL}$) whenever φ is compound. For example, $((p \rightarrow p) \rightarrow x) \wedge z \rightarrow x$ is provable.

Exhaustive tests over all formulas with at most four binary connectives in $\rightarrow, \wedge, \vee$ and three variables give the following. There are 413 formulas minimal in $\mathbf{FL}_e$, up to renaming. For 164 of them $S(\alpha)$ is provable, and in all 164 cases it is minimal, as Theorem 1.1 predicts. φ^+ is provable in all 413 cases and minimal in none. Wrapping only the links strictly below an additive occurrence gives a provable formula in all 413 cases, but a minimal one in only 255. For $\alpha = p \vee (q \rightarrow q)$, for instance, the wrapper can be removed because the $\vee$ -right rule selects one branch. So a minimality-preserving translation over $\mathbf{FL}_e$, if one exists, must adapt the wrappers to the actual proof. Whether such a translation exists, and the non-commutative case $\mathbf{FL}$, are left open.

6.3 Remarks on the proofs

Remark 6.4. Corollary 5.5 replaces [14, Props. 3.1–3.2], whose proofs there are sketched, and it does not need simple formulas.

Remark 6.5. In Lemma 3.4(i), linear λ -terms can be given instead of sequent derivations. Without exchange the premises of $S(\alpha)$ cannot be reordered; see Section 6.1.

6.4 Top-level premises of order 3

For $F = A_1 \rightarrow \cdots \rightarrow A_n \rightarrow r$ with r a variable, we call $A_1, \dots, A_n$ the *top-level premises* of F . The next proposition shows that an order-4 counter-example needs at least two top-level premises of order 3. This explains, in part, why $S(G')$ is long.

Proposition 6.6. *Let $F = A_1 \rightarrow \cdots \rightarrow A_n \rightarrow r \in \mathbf{CL}$ with r a variable and $\text{Ord}(F) \leq 4$. If at most one A_i has order 3, then $F \in \mathbf{IL}$ or $\mathbf{IL} + F = \mathbf{CL}$. Minimality of F is not assumed.*

Proof. A formula of order ≤ 2 has the form $p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$ with variables p_i, q ($k \geq 0$). We call it a *Horn clause* and read it as the rule “from $p_1, \dots, p_k$ infer q ”. Since $\text{Ord}(F) \leq 4$, every A_i has order ≤ 3 . Recall that premises can be permuted in $\mathbf{IL}$.

Let Γ be the set of top-level premises of order ≤ 2 . For a set X of variables, let $C(X)$ be the closure of X under the rules in Γ . We identify a set of variables with the valuation making exactly those variables true. Then:

(H1) $C(X)$ satisfies Γ ;

(H2) if $q \in C(X)$, then $\Gamma, X \vdash_{\mathbf{IL}} q$ (modus ponens along the closure). In particular, $q \in C(\{p_1, \dots, p_k\})$ implies $\Gamma \vdash_{\mathbf{IL}} p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$.

No premise has order 3. Then F is $\Gamma \rightarrow r$. By (H1) and $F \in \mathbf{CL}$, we get $r \in C(\emptyset)$, so $F \in \mathbf{IL}$ by (H2).

Exactly one premise, $A = B_1 \rightarrow \cdots \rightarrow B_m \rightarrow s$, has order 3. Then each B_j has order ≤ 2 . Assume $F \notin \mathbf{IL}$. Put $C_0 = C(\emptyset)$.

- $r \notin C_0$, since otherwise $\Gamma \vdash_{\mathbf{IL}} r$ by (H2).

- Some B_j is not derivable from Γ in **IL**. Otherwise $\Gamma, A \vdash_{\mathbf{IL}} s$. Moreover, the valuation $C(\{s\})$ satisfies Γ and A , so $r \in C(\{s\})$ because $F \in \mathbf{CL}$. Then $\Gamma, s \vdash_{\mathbf{IL}} r$ by (H2), and $F \in \mathbf{IL}$.

Fix such a $B_j = p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$, and put $C_1 = C(\{p_1, \dots, p_k\})$. Then $C_0 \subseteq C_1$, and $q \notin C_1$ by (H2). The valuation C_1 falsifies B_j , hence satisfies A . It also satisfies Γ by (H1). Since $F \in \mathbf{CL}$, we get $r \in C_1$.

Let $H_3 = \{0, \frac{1}{2}, 1\}$ be the three-element Gödel algebra, with $x \Rightarrow y = 1$ if $x \leq y$ and $x \Rightarrow y = y$ otherwise. Define $v(t) = 1$ if $t \in C_0$, $v(t) = \frac{1}{2}$ if $t \in C_1 \setminus C_0$, and $v(t) = 0$ otherwise. A Horn clause $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ has value 1 under v iff $\min_i v(p_i) \leq v(q)$ (the minimum of an empty set is 1).

- Every clause in Γ has value 1, since C_0 and C_1 are closed under the rules in Γ .
- B_j has value 0, since $v(p_i) \geq \frac{1}{2}$ for all i and $v(q) = 0$. Hence A has value 1.
- $v(r) = \frac{1}{2}$, since $r \in C_1 \setminus C_0$.

So F has value $\frac{1}{2}$, and $H_3 \not\models F$. By Jankov's criterion [11, Theorem 1] (see [14, Sect. 3.2]), an implicational $F \in \mathbf{CL} \setminus \mathbf{IL}$ with $H_3 \not\models F$ satisfies $\mathbf{IL} + F = \mathbf{CL}$. $\square$

Remark 6.7. In $S(\alpha)$, a positive link $b_u \rightarrow x_u$ has order 3 and a negative link $x_u \rightarrow b_u$ has order 2. So the top-level premises of order 3 in $S(\alpha)$ are exactly the links of the positive implication occurrences of α . Hence, by Proposition 6.6, if $\alpha \in \mathbf{CL}$ has at most one positive implication occurrence, then $\mathbf{IL} + \alpha = \mathbf{IL} + S(\alpha)$ is **IL** or **CL**.

$S(G')$ has 11 positive links: $E_1, E_3, E_5, E_7, E_9, E_{11}, E_{12}, E_{14}, E_{15}, E_{17}, E_{20}$. By Proposition 6.6, a **CL**-minimal axiom of order 4 for a proper intermediate logic has at least two top-level premises of order 3. The least possible number is open.

7 Formalization

[Plot: Half a page.]

File `lean/StatmanMinimality.lean` (about 1300 lines, implicational language): Lean 4 core only, no Mathlib, no `sorry`, no `native_decide`. `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. Theorem 1.1 is formalized in exactly the generality stated: a logic is a predicate `Th` with `PreLogic Th` (closed under substitution and modus ponens, containing every instance $\gamma \rightarrow \gamma$ of I), and `statman_minimalIn : PreLogic Th  $\rightarrow$  ( $\forall q, 0 \leq q \leq N$ )  $\rightarrow$  MinimalIn Th A  $\rightarrow$  Th (statman A N)  $\rightarrow$  MinimalIn Th (statman A N)`. The classical instance `statman_minimal` uses `logic_valid` and `statman_valid`. Lemma 3.4(i) and Corollary 3.6 are formalized for every implicational logic containing **BCI** (`BCIExt Th`: closed under substitution and modus ponens, proving all instances of B, C, I): `BCIExt.statman_imp : Th (A  $\rightarrow$  statman A N), BCIExt.statman_iff : Th A  $\leftrightarrow$  Th (statman A N)`, and `BCIExt.statman_min` (Corollary 1.2 in the implicational language, without extra hypotheses). Also formalized: `statman_ord` (order ≤ 4) and `statman_Gprime` (Corollary 4.2: the displayed $S(G')$ equals `statman Gprime` 8). Not formalized: the full language (Lemma 3.4(ii),(iii)). Correspondence: `reduction` and `strict` = Corollary 5.5; `decode/decode_node` = Lemma 5.8 (organized node by node rather than by Lemma 5.7).

8 Concluding Remarks

We showed that the polarized Statman translation preserves minimality with respect to the substitution preorder in any logic that contains the axiom I and $S(\alpha)$. As a consequence, **CL**-minimal axiomatizations of implicational logics over **BCI** can be taken of order 4, and [14, Open Problem 2] has an affirmative answer.

Open problems:

1. Which intermediate logics are axiomatizable by **CL**-minimal formulas [14, Open Problem 1]? By Corollary 1.3 it suffices to consider formulas of order 4.
2. Is there a minimality-preserving translation over **FL_e** with additive connectives, or over the non-commutative **FL** (Section 6.2)?
3. Does the translation (possibly with reordered premises) preserve minimality over ticket entailment $\mathbf{T}_{\rightarrow} = \mathbf{BB'IW}$ (Section 6.1)?
4. What is the least number of top-level premises of order 3 in a **CL**-minimal order-4 axiom of a proper intermediate logic? It is at least 2 (Proposition 6.6), and $S(G')$ has 11.

Acknowledgements. [\[Plot: JSPS KAKENHI etc.\]](#)

A The formula $S(G')$

The implication occurrences of G' are numbered $1, \dots, 22$ in pre-order, and x_i is the extension variable of occurrence i . $S(G')$ lists the links in the order of the table (post-order), followed by x_1 . The last column gives the subformula G'_u at occurrence i , so that the decoding substitution of Lemma 3.5 is $x_i \mapsto G'_u$. The table is generated by `paper/gen_SGprime.py`.

Table 1: The links of $S(G')$.

i	pol.	E_i	G'_u
E_2	—	$x_2 \rightarrow a \rightarrow a'$	$a \rightarrow a'$
E_4	—	$x_4 \rightarrow b' \rightarrow b$	$b' \rightarrow b$
E_6	—	$x_6 \rightarrow c'' \rightarrow c'$	$c'' \rightarrow c'$
E_8	—	$x_8 \rightarrow c''' \rightarrow c'$	$c''' \rightarrow c'$
E_{11}	+	$(a \rightarrow b) \rightarrow x_{11}$	$a \rightarrow b$
E_{10}	—	$x_{10} \rightarrow x_{11} \rightarrow c''$	$(a \rightarrow b) \rightarrow c''$
E_{14}	+	$(b' \rightarrow a') \rightarrow x_{14}$	$b' \rightarrow a'$
E_{13}	—	$x_{13} \rightarrow x_{14} \rightarrow c'''$	$(b' \rightarrow a') \rightarrow c'''$
E_{18}	—	$x_{18} \rightarrow a' \rightarrow b'$	$a' \rightarrow b'$
E_{17}	+	$(x_{18} \rightarrow c'') \rightarrow x_{17}$	$(a' \rightarrow b') \rightarrow c''$
E_{21}	—	$x_{21} \rightarrow b \rightarrow a$	$b \rightarrow a$
E_{20}	+	$(x_{21} \rightarrow c''') \rightarrow x_{20}$	$(b \rightarrow a) \rightarrow c'''$
E_{22}	—	$x_{22} \rightarrow c' \rightarrow c$	$c' \rightarrow c$
E_{19}	—	$x_{19} \rightarrow x_{20} \rightarrow x_{22}$	$((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$
E_{16}	—	$x_{16} \rightarrow x_{17} \rightarrow x_{19}$	$((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$
E_{15}	+	$(x_{16} \rightarrow c) \rightarrow x_{15}$	$((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$

i	pol.	E_i	G'_u
E_{12}	+	$(x_{13} \rightarrow x_{15}) \rightarrow x_{12}$	$((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_9	+	$(x_{10} \rightarrow x_{12}) \rightarrow x_9$	$((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_7	+	$(x_8 \rightarrow x_9) \rightarrow x_7$	$(c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_5	+	$(x_6 \rightarrow x_7) \rightarrow x_5$	$(c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_3	+	$(x_4 \rightarrow x_5) \rightarrow x_3$	$(b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_1	+	$(x_2 \rightarrow x_3) \rightarrow x_1$	$(a \rightarrow a') \rightarrow (b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$

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